

POCKETSMART AI

Project Report

Abstract

PocketSmart AI is a Generative AI-powered web application developed to assist users with budget planning and personalized recommendations. The system allows users to provide information such as budget, room type, preferred style, and required items. The application processes the information and uses Generative AI to generate a personalized plan.

Introduction

In today's digital environment, users have access to many choices but may find it difficult to select suitable options within a specific budget. PocketSmart AI addresses this problem by combining web technologies with Generative AI.

Existing System

In a traditional approach, users manually search for options, compare information, calculate costs, and create their own plans. This process can require significant time and effort.

Proposed System

PocketSmart AI provides an intelligent web-based solution where users enter their requirements and receive an AI-generated plan and recommendations.

Objectives

The main objective is to develop an easy-to-use AI-powered assistant that can understand user requirements and provide personalized planning assistance.

Technologies Used

The project uses Python, FastAPI, HTML, CSS, JavaScript, SQLite, SQLAlchemy, and a Generative AI API.

Implementation

The application consists of authentication, planner, database, AI service, recommendation, and result display modules.

Generative AI Integration

The application converts user requirements into an appropriate prompt and sends the request to the Generative AI service. The returned response is processed and displayed to the user as a personalized plan.

Testing

The application is tested using functional, validation, integration, and error-handling test cases.

Result

The completed system provides an interface where users can enter their requirements and receive an AI-generated plan and recommendations.

Future Enhancement

Future versions can include advanced budget tracking, product price comparison, personalized history, additional planning categories, improved recommendation capabilities, and mobile application support.

Conclusion

PocketSmart AI demonstrates how Generative AI can be integrated with a web application to provide personalized planning assistance. The project combines artificial intelligence, backend development, database management, and user interface design into a single practical application.